Exam 2 – Math 110.

*You may use a basic calculator and formula sheets. Your sheets must contain formulas only, no examples. Phones and all other smart devices are forbidden. Any AI usage will result in an automatic F for the exam. Show **ALL** your work for credits.*

1. Solve each quadratic by factoring or the quadratic formula.

   a. $x^2 = 9$

   $x^2 - 9 = 0$

   $x \quad 3$

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   $(x-3)(x+3) = 0$

   $\boxed{x = \pm 3}$

   b. $2x^2 + 12x + 18 = 0$

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   $a = 2 \quad b = 12 \quad c = 18$

   $(12)^2 - 4(2)(18)$

   $= 0$

   $\boxed{x = -3}$

   $\frac{-12 \pm 0}{4}$

   $\frac{-12 + 0}{4} = -3$ $\quad$ $\frac{-12 - 0}{4} = -3$

   c. $3x^2 + 6x - 9 = 0$

   $3x^2 + 6x - 9 = 0$

   $a = 3 \quad b = 6 \quad c = -9$

   $(6)^2 - 4(3)(-9)$

   $= 144$

   $\frac{-6 \pm \sqrt{144}}{6}$ $\quad$ $\frac{-6 + 12}{6} = 1$ $\boxed{x = 1}$

   $\frac{-6 - 12}{6} = -3$ $\boxed{x = -3}$

   d. $x^2 + 3x = 2x + 12$

   $x^2 + 3x - 2x - 12 = 0$

   $x^2 + 1 - 12 = 0$

   $a = 1 \quad b = 1 \quad c = -12$

   $(1)^2 - 4(1)(-12)$

   $= 49$

   $\frac{-1 \pm \sqrt{49}}{2}$ $\quad$ $\frac{-1 + 7}{2} = 3$ $\boxed{x = 3}$

   $\frac{-1 - 7}{2} = -4$ $\boxed{x = -4}$

2. Graph of the quadratic functions. Label the vertex and another point.

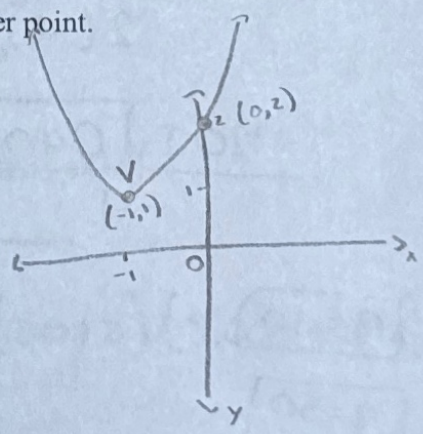
   a. $y = x^2 + 2x + 2$ $\quad a = 1 \quad b = 2 \quad c = 2$

   $V = \left(-\frac{2}{2(1)}, \frac{4(1)(2) - (2)^2}{4(1)}\right)$

   $V = (-1, 1)$

   Another Point

   $x = 0 \quad y = 2$



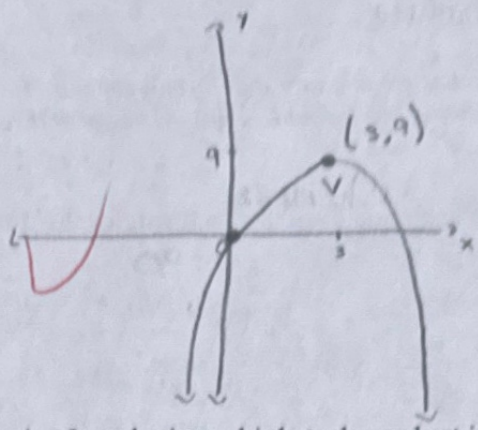
b. $y = -x^2 + 6x$ $a = -1$ $b = 6$ $c = 0$

$$V = \left(-\frac{6}{2(-1)}, \frac{4(-1)(0) - (6)^2}{4(-1)}\right)$$

$V = (3, 9)$

Another Point

$x = 0$ $y = 0$



3. Suppose that in a monopoly market the total cost per week of producing a high-tech product is given by $C = 1000 + 40q + q^2$. Suppose further that the sales price function for this product is $p = 160 - q$.

a. Find the revenue function in term of q. $R = P \times Q$

$R = (160 - q)(q)$

$\boxed{R = -q^2 + 160q}$

b. Find the number of units that will **maximize the revenue.**

$R = -q^2 + 160q$ $a = -1$ $b = 160$ $c = 0$

$q = \frac{-b}{2a}$ Maximize when $q = -\frac{160}{2(-1)} = 80$

c. Find the profit function Profit = R - C

$P = -q^2 + 160q - (1{,}000 + 40q + q^2)$

$= -q^2 + 160q - 1{,}000 - 40q - q^2$

$\boxed{\text{Profit} = -2q^2 + 120q - 1{,}000}$

d. Find the number of units that will give **break-even** for the product

$-2q^2 + 120q - 1{,}000 = 0$ $a = -2$ $b = 120$ $c = -1{,}000$

$$\frac{-120 \pm \sqrt{(120)^2 - 4(-2)(-1{,}000)}}{2(-2)}$$

$$\frac{-120 \pm \sqrt{6400}}{-4}$$

$$\frac{-120 + 80}{-4} = 10$$

$$\frac{-120 - 80}{-4} = 50$$

$\boxed{q = 10}$

$\boxed{q = 50}$

e. Find the number of products need to maximize the profit. $q = -\frac{b}{2a}$

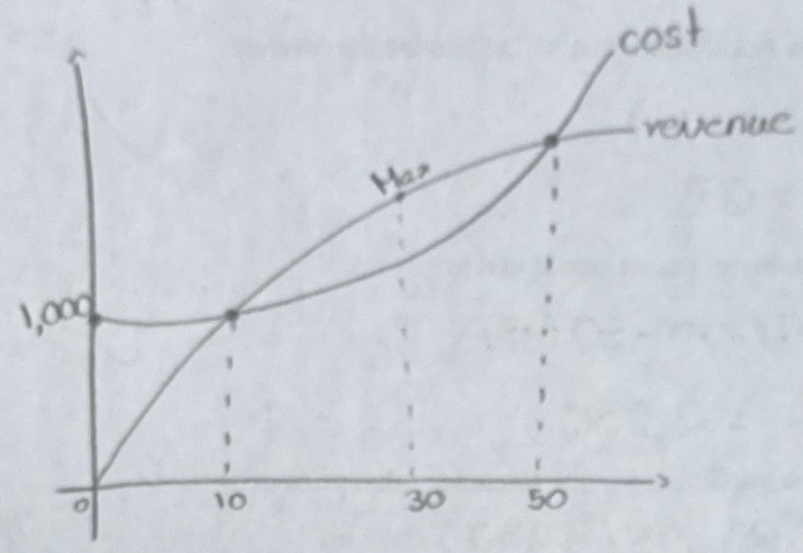
Profit $= -2q^2 + 120q - 1{,}000$ $a = -2$ $b = 120$ $c = -1{,}000$

Maximize when $q = \frac{-120}{2(-2)} = 30$

$-2(30)^2 + 120(30) - 1{,}000$
$= 800$

$q = 30$

f. Graph the revenue function and the cost function label the break-even points, fixed cost, and the maximized profit point.



4. On a certain route, an airline carries 4000 passengers per month, each paying $50. A market survey indicates that for each $1 increase in the ticket price, the airline will lose 50 passengers.

a. What is the airline's current revenue?

$R = 50 \times 4{,}000$

$\boxed{R = 200{,}000}$

b. Create an income (revenue) function if "x" is defined as the number of $1 price increases

	old	new
price	50	$50 + x$
quantity	4,000	$4{,}000 - 50x$
revenue	$50 \times 4{,}000$	$(50+x)(4{,}000-50x)$

New Revenue function $= (50+x)(4{,}000+50x)$

c. Find the number of \$1 price increases that will maximize the revenue.

$R = (50 + x)(4{,}000 - 50x)$

$= 200{,}000 - 2{,}500x + 4{,}000x - 50x^2$

$-50x^2 + 1{,}500x + 200{,}000 = 0$

$a = -50 \quad b = 1{,}500 \quad c = 200{,}000$

Maximize when $x = \dfrac{-1{,}500}{2(-50)} = 15$

d. Find the new ticket price (that will maximize the revenue)

$50 + (15)$

$= 65$

e. Find the number of passengers at that price in d.

$4{,}000 - 50(15)$

$= 3{,}250$

f. Find the new maximum income (income at that price in d)

$R = 65(3{,}250)$

$\boxed{R = 211{,}250}$

5. If the supply function for a commodity is given by $p = q^2 + 4q$ and the demand function is given by $p = 96 - q^2$, find the point of market equilibrium (Supply equals Demands).

$q^2 + 4q = 96 - q^2$

$q^2 + q^2 + 4q - 96 = 0$

$2q^2 + 4q - 96 = 0$

$a = 2 \quad b = 4 \quad c = -96$

$\dfrac{-4 \pm \sqrt{(4)^2 - 4(2)(-96)}}{2(2)}$

$\dfrac{-4 \pm \sqrt{784}}{4}$

$\dfrac{-4 + 28}{4} = 6$

$\boxed{q = 6}$

$\dfrac{-4 - 28}{4} = -8$

$p = 96 - (6)^2$

$P = 96 - 36$

$\boxed{P = 60}$